MANAGING PROCESSES AND OPTIMIZING PERFORMANCE

PROCESS PRIORITY AND NICE VALUES

What is process priority

1. Every process in Linux has a priority that decides how much CPU time it gets.
2. Higher priority → gets CPU first
3. Lower priority → waits longer
4. **nice and renice** are commands used to manage process priorities.
5. Priorities are as follows,
   - **-20 ---------> most priority (least nice, more CPU)**
   - **+19 --------> least priority**
   - **0 ----------> Default priority (most nice, less CPU)**

1.nice (start process - start process with priority)

1. Used to start a process with a specific nice value
2. **Syntax:** nice -n <value> command
3. **Example:**
   nice -n 10 sleep 100 ----------> { Starts process with lower priority }

2.renice (modify process - change running process priority)

1. Used to change priority of an already running process
2. **Syntax:** renice -n <value> -p <PID>
3. **Example:**
   renice -n -5 -p 1234 ----------> { Gives higher priority to process with PID 1234 }

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EXAMPLE: 1

1.sleep 200
ctrl + z

2.ps

```
  PID    TTY        TIME CMD
 1044    pts/2      00:00:00 sleep
```

3.ps -l 1044

```
F S   UID    PID    PPID    C   PRI   NI   ADDR   SZ     WCHAN     TTY       TIME CMD
0 T    0     1044    981     0   80    0     -     803    do_sig     pts/2     0:00 sleep 200
```

4.renice -n -5 -p 1044
{ 1044 (process ID) / old priority 0, new priority -5 }

5. ps -l 1044

F	S	UID	PID	PPID	C	PRI	NI	ADDR	SZ	WCHAN	TTY	TIME CMD
0	T	0	1044	981	0	75	-5	-	803	do_sig	pts/2	0:00 sleep 200

EXAMPLE: 1

1.nice -n -15 sleep 200

ctrl + z

2.ps

PID	TTY	TIME CMD
931	pts/2	00:00:00 sleep

3.ps -l 931

F	S	UID	PID	PPID	C	PRI	NI	ADDR	SZ	WCHAN	TTY	TIME CMD
0	T	0	931	981	0	65	-15	-	803	do_sig	pts/2	0:00 sleep 200

4. renice -n -5 -p 931

5.ps -l 931

F	S	UID	PID	PPID	C	PRI	NI	ADDR	SZ	WCHAN	TTY	TIME CMD
0	T	0	931	981	0	75	-5	-	803	do_sig	pts/2	0:00 sleep 200

Note:

1. Only root user can set negative values (-20 to -1)
2. Normal users can set 0 to +20 only
3. Lower nice value = higher priority

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